

Continuous Families of Contractions with Discontinuous Fixed Points: Dynamic Programming and Production Networks

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Abstract

A continuously parameterized family of contractions need not have continuously varying fixed points. An explicit example in ℓ^2 , built from finite nilpotent shifts, motivates two classical remedies: a contraction bound holding locally in the parameter, or a compact neighborhood of the limiting fixed point. We connect these criteria to discounted dynamic programming and production networks. Ordinary scalar-discount Bellman operators supply the required local modulus control. In contrast, the same finite shifts on ℓ^1 produce acyclic input–output networks whose technologies converge strongly and whose final demands converge in norm, while aggregate gross output has a persistent error. Every spectral radius is zero. For nonnegative column-substochastic matrices with N sectors and spectral radius at most $r < 1$, we derive the sharp multiplier bound $N/(1-r)$. A common finite-round loss of intermediate requirements gives a bound independent of sector count, restoring continuity and certifying errors in approximate production plans.

Keywords: fixed points; contractions; dynamic programming; production networks; input–output analysis.

JEL: C61, C62, D57.

1 Motivation

In dynamic programming, the value function is a fixed point of a Bellman operator (Stokey et al., 1989). In input–output analysis, gross production solves $x = d + Ax$, where d is final demand and A records direct input requirements (Miller and Blair, 2022). Both settings require understanding how a fixed point changes with the primitives. Existence and uniqueness at each parameter do not, by themselves, ensure continuity across parameters. The counterexample below makes this failure explicit: every operator is a contraction and the family is jointly continuous, yet its fixed points stay at distance one from the limiting fixed point.

The proof of Theorem 7.18 in de la Fuente (2000, pp. 88–89) uses a common contraction modulus absent from its assumptions, as noted by Shannon (n.d., item 7). The underlying mathematical issue is classical: Nadler (1968) established both the infinite-dimensional failure and continuity under local compactness. We present the failure through finite nilpotent shifts and give short proofs of two classical continuity criteria. Discounted Bellman operators illustrate the local contraction

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criterion. The same shifts then describe production chains in which a spectral gap ensures existence but does not uniformly control accumulated requirements as the sector list expands. We derive a sharp bound that retains sector count and a finite-round condition that removes this dependence. The common issue is the amplification of small perturbations in the fixed-point equation; bounded inverses provide the corresponding stability condition for affine models (Zhao et al., 2025).

2 The continuity claim and its missing hypothesis

In Sections 2–5, (X, d) is a nonempty complete metric space, Ω is a metric parameter space, and $T_\alpha : X \rightarrow X$ is a contraction with modulus $\beta_\alpha < 1$. Its unique fixed point is z_α . Parameter continuity means $T_\alpha x \rightarrow T_{\alpha_0} x$ in X for every fixed $x \in X$ whenever $\alpha \rightarrow \alpha_0$. Section 6 specializes to affine maps on ℓ^1 and also treats nonnegative matrices whose column sums may equal one.

The claim under examination. Theorem 7.18 of de la Fuente (2000, pp. 88–89) asserts that completeness of X , continuity of $\alpha \mapsto T_\alpha x$ for every fixed x , and contractivity of each T_α imply continuity of the fixed-point map $\alpha \mapsto z_\alpha$. The contraction factor may vary with the parameter. This assertion is false in general, even if $(x, \alpha) \mapsto T_\alpha x$ is jointly continuous.

Fix α_0 and write $z_0 = z_{\alpha_0}$ and $\beta_0 = \beta_{\alpha_0}$. The triangle inequality and the fixed-point equations give

$$d(z_\alpha, z_0) \leq \beta_\alpha d(z_\alpha, z_0) + d(T_\alpha z_0, T_{\alpha_0} z_0), \quad d(z_\alpha, z_0) \leq \frac{d(T_\alpha z_0, T_{\alpha_0} z_0)}{1 - \beta_\alpha}. \quad (1)$$

The numerator tends to zero. The inference that the quotient does too needs an additional argument; dividing by $1 - \beta_\alpha > 0$ is itself valid. A common bound $\beta_\alpha \leq q < 1$ makes the denominator uniformly positive and proves continuity, but individual contractivity alone does not supply that bound. The next section gives an explicit failure in ℓ^2 ; Section 4 states and proves two sufficient conditions.

3 A jointly continuous counterexample

Take $X = \ell^2$ with its usual orthonormal basis (e_i) and $\Omega = \{0\} \cup \{1/n : n \geq 2\}$. Write $m_n = 2^n$ and define the finite shift

$$S_n e_i = \begin{cases} e_{i+1}, & m_n \leq i < m_n + n, \\ 0, & \text{otherwise.} \end{cases}$$

The blocks $\{m_n, \dots, m_n + n\}$ are disjoint, $\|S_n\| = 1$, and $S_n^{n+1} = 0$. Set

$$q_n = 1 - \frac{1}{n}, \quad c_n = \left(\sum_{k=0}^n q_n^{2k} \right)^{-1/2}, \quad T_{1/n} x = q_n S_n x + c_n e_{m_n}, \quad T_0 x = 0. \quad (2)$$

Proposition 1. *The family in (2) is jointly continuous, and each T_α is a contraction. Nevertheless,*

$$z_{1/n} = c_n \sum_{k=0}^n q_n^k e_{m_n+k}, \quad \|z_{1/n}\|_2 = 1, \quad z_0 = 0.$$

Thus the fixed-point map is discontinuous at zero.

Proof. The contraction moduli are $q_n < 1$ and 0. The geometric-series identity gives

$$c_n^2 = \frac{1 - q_n^2}{1 - q_n^{2(n+1)}} \sim \frac{2}{(1 - e^{-2})n} \longrightarrow 0.$$

For each fixed $x \in \ell^2$,

$$\|S_n x\|_2^2 = \sum_{i=m_n}^{m_n+n-1} |x_i|^2 \leq \sum_{i \geq m_n} |x_i|^2 \longrightarrow 0.$$

Hence $T_{1/n} x \rightarrow 0$. Moreover, $\|T_{1/n} y - T_0 x\|_2 \leq \|y - x\|_2 + \|S_n x\|_2 + c_n$, proving joint continuity at $(x, 0)$; the other parameters are isolated. Nilpotence yields $(I - q_n S_n)^{-1} = \sum_{k=0}^n q_n^k S_n^k$, so the displayed expression for $z_{1/n}$ follows from the fixed-point equation. Its squared norm is $c_n^2 \sum_{k=0}^n q_n^{2k} = 1$. $\square$

A vanishing forcing term is amplified along increasingly long finite chains. The fixed points have disjoint supports, so $\|z_{1/n} - z_{1/m}\|_2 = \sqrt{2}$ for $n \neq m$: boundedness supplies no convergent subsequence. Although each linear part is nilpotent, the nearby contraction moduli approach one. This is the feature addressed by the first criterion below. The example also separates pointwise convergence from uniform convergence on bounded sets: evaluating at $z_{1/n}$ gives

$$\sup_{\|x\|_2 \leq 1} \|T_{1/n} x - T_0 x\|_2 \geq 1.$$

4 Two continuity criteria

The missing control in (1) can come from the contraction moduli near α_0 or from compactness near z_0 .

Theorem 1. *Assume parameter continuity. Then $z_\alpha \rightarrow z_0$ as $\alpha \rightarrow \alpha_0$ if either of the following holds:*

- (i) *There are a neighborhood U of α_0 and $q < 1$ such that $\beta_\alpha \leq q$ for every $\alpha \in U$.*
- (ii) *z_0 has a compact neighborhood in X .*

In particular, (ii) applies whenever X is a closed subset of a finite-dimensional normed space, without a common contraction modulus. It also applies to any proper metric space, meaning one in which closed bounded sets are compact.

Proof. Under (i), the denominator in (1) is at least $1 - q > 0$.

For (ii), the local compactness argument is due to Nadler (1968, Theorem 2). Fix any sufficiently small $r > 0$ so that $K = \{x \in X : d(x, z_0) \leq r\}$ is compact. All the operators are 1-Lipschitz. A finite ε -net in K , together with parameter continuity at its finitely many points, therefore gives

$$\delta_\alpha := \sup_{x \in K} d(T_\alpha x, T_{\alpha_0} x) \longrightarrow 0.$$

Indeed, the error at any point of K is at most 2ε plus the largest error on the net. For α sufficiently close to α_0 , $\delta_\alpha \leq (1 - \beta_0)r$, and thus

$$d(T_\alpha x, z_0) \leq \delta_\alpha + \beta_0 r \leq r \quad (x \in K).$$

Consequently $T_\alpha(K) \subseteq K$. Banach's theorem on the complete space K , and uniqueness in X , imply $z_\alpha \in K$. Since r can be arbitrarily small, $z_\alpha \rightarrow z_0$. $\square$

At a nonisolated parameter, condition (i) is equivalent to $\limsup_{\alpha \rightarrow \alpha_0} \beta_\alpha < 1$. Upper semicontinuity of the moduli is sufficient and strictly stronger than this local bound: nearby moduli may have limit superior below one while exceeding β_0 . A bound uniform over all of Ω is unnecessary. Neither criterion is necessary for a particular family: removing the forcing term from (2) leaves the same moduli on ℓ^2 but makes every fixed point zero.

5 Discounted dynamic programming

Let S be a nonempty measurable state space and X a closed linear subspace of bounded measurable functions on S , with the supremum norm, containing the constants. Feasible action sets $\Gamma(s)$ are nonempty, rewards are bounded by $R = \sup_{s,a} |r(s,a)| < \infty$, and $Q(\cdot \mid s, a)$ is a transition probability. Hold these primitives fixed and assume the Bellman operator

$$(T_\beta v)(s) = \sup_{a \in \Gamma(s)} \left\{ r(s, a) + \beta \int v(s') Q(ds' \mid s, a) \right\}, \quad \beta \in (0, 1), \quad (3)$$

maps X into itself. This last assumption includes the needed measurability of the supremum.

The usual contraction argument (Blackwell, 1965; Stokey et al., 1989) gives

$$\|T_\beta v - T_\beta w\|_\infty \leq \beta \|v - w\|_\infty, \quad \|T_\beta v - T_\gamma v\|_\infty \leq |\beta - \gamma| \|v\|_\infty.$$

The modulus is exactly β , because $T_\beta(v + c\mathbf{1}) = T_\beta v + \beta c\mathbf{1}$ for every constant c . Although $\sup_{\beta \in (0,1)} \text{Lip}(T_\beta) = 1$, the moduli are locally bounded away from one at every interior discount factor. Theorem 1(i) therefore gives continuity of the value function $\beta \mapsto v_\beta$ in supremum norm. More explicitly, $\|v_\gamma\|_\infty \leq R/(1 - \gamma)$ and (1) imply

$$\|v_\beta - v_\gamma\|_\infty \leq \frac{R|\beta - \gamma|}{(1 - \beta)(1 - \gamma)}. \quad (4)$$

Equality holds in the one-state problem with constant reward R , so the bound also records the loss of uniform control as discounting vanishes.

The same reasoning covers a parameter vector. If $T_\alpha : X \rightarrow X$ is monotone and satisfies exact scalar discounting, $T_\alpha(v + c\mathbf{1}) = T_\alpha v + \beta_\alpha c\mathbf{1}$, with $\beta_\alpha \in [0, 1]$, monotonicity gives $\text{Lip}(T_\alpha) \leq \beta_\alpha$ and constants give equality. Furthermore,

$$\beta_\alpha \mathbf{1} = T_\alpha \mathbf{1} - T_\alpha 0.$$

If $\alpha \mapsto T_\alpha v$ is continuous in supremum norm for every $v \in X$, evaluating at $\mathbf{1}$ and 0 shows that β_α is continuous. The local criterion then gives continuity of the fixed points. Thus ordinary scalar-discount Bellman operators themselves supply the modulus control missing from the abstract statement.

6 Production networks and accumulated requirements

The local bound in Section 5 controls how perturbations propagate through future rewards. Production networks pose the analogous question for successive rounds of intermediate inputs. Use the common sector space ℓ^1 , with $\|x\|_1 = \sum_i |x_i|$, and record output at fixed accounting prices. A finite network is embedded by setting unused coordinates to zero. In the equation $x = d + Ax$, the nonnegative entry A_{ij} is the value of input i required per unit of output value in sector j . We study column-substochastic operators, $\|A\|_1 \leq 1$; in this normalization each column's total intermediate requirements do not exceed one.

Write $\text{spr}(A)$ for the spectral radius. If $\text{spr}(A) < 1$, the production plan is $x = R_A d$, where $R_A = (I - A)^{-1} = \sum_{k \geq 0} A^k$ is the requirement multiplier. For two productive economies, the fixed-point equations give

$$x_n - x_0 = (I - A_n)^{-1} \{d_n - d_0 + (A_n - A_0)x_0\}. \quad (5)$$

This is the affine counterpart of (1). Strong convergence $A_n y \rightarrow A_0 y$ for each fixed y , together with $d_n \rightarrow d_0$, makes the expression in braces tend to zero. A common bound on the inverses then ensures continuity. The next example shows why a common spectral gap alone does not provide such a bound as the number of sectors increases.

6.1 The same finite chains in the production norm

Use the shifts S_n of Section 3, now acting on ℓ^1 , and set

$$A_n = q_n S_n, \quad H_n = \sum_{k=0}^n q_n^k = n(1 - q_n^{n+1}), \quad d_n = e_1 + H_n^{-1} e_{m_n}. \quad (6)$$

The limiting economy is $A_0 = 0$, $d_0 = e_1$ and $x_0 = e_1$. The demand increment H_n^{-1} normalizes the chain in ℓ^1 ; the coefficient c_n in (2) instead normalizes it in ℓ^2 . The isolated first sector supplies a common baseline production level.

Proposition 2. *Each A_n is nonnegative, $\|A_n\|_1 = q_n < 1$ and $\text{spr}(A_n) = 0$. Moreover, $A_n y \rightarrow 0$ for every fixed $y \in \ell^1$ and $d_n \rightarrow d_0$, but*

$$x_n = e_1 + H_n^{-1} \sum_{k=0}^n q_n^k e_{m_n+k}, \quad \|x_n - x_0\|_1 = 1.$$

The multiplier norm is $\|(I - A_n)^{-1}\|_1 = H_n \sim (1 - e^{-1})n$.

Proof. The matrix is nilpotent, $A_n^{n+1} = 0$, with largest column sum q_n . For fixed y ,

$$\|A_n y\|_1 \leq \sum_{i \geq m_n} |y_i| \rightarrow 0.$$

Also $q_n^{n+1} \rightarrow e^{-1}$, so $H_n^{-1} \rightarrow 0$. The finite inverse $\sum_{k=0}^n A_n^k$ gives the stated solution, whose added components sum to one. The largest column sum of this inverse is H_n , attained at the beginning of the chain; columns outside the chain sum to one. $\square$

Thus $x \mapsto d_n + A_n x$ is another jointly continuous family of contractions with discontinuous fixed points, by the same 1-Lipschitz argument used in Proposition 1. Every fixed finite list of sectors eventually records the limiting production plan exactly. Every fixed number of rounds also misses the extra production asymptotically: through round K , including direct demand as round zero, the fraction captured is

$$\frac{\sum_{k=0}^{\min\{K,n\}} q_n^k}{H_n} = \frac{1 - q_n^{\min\{K,n\}+1}}{1 - q_n^{n+1}} \rightarrow 0. \quad (7)$$

Taking all rounds first gives one. No cycle is needed, and every spectral radius is zero. The example changes technology on a common sector list; it is not an exact truncation scheme for one fixed operator.

Aggregate gross output is two in each approximating economy and one in the limit. This does not establish a discontinuity in value added. With $a_{n,j} = 1 - \sum_i (A_n)_{ij}$, the accounting identity is

$$\sum_j a_{n,j} (x_n)_j = \sum_j (d_n)_j = 1 + H_n^{-1} \rightarrow 1. \quad (8)$$

Gross production counts intermediate stages repeatedly, and that is the quantity whose robustness fails here. Nor does the example contradict Theorem 1(ii): the active sector count grows, so the production plans do not lie in a common finite-dimensional space. Compactness at a fixed dimension does not supply a bound uniform over expanding sector lists.

6.2 How accuracy depends on sector count

For a fixed number of sectors, a spectral gap does yield a quantitative bound. The following result also allows column sums equal to one, extending beyond the strict one-round contractions of the example.

Proposition 3. *If $A \geq 0$ is $N \times N$, $\|A\|_1 \leq 1$, and $\text{spr}(A) \leq r < 1$ with $r \geq 0$, then*

$$\|(I - A)^{-1}\|_1 \leq \frac{N}{1 - r}. \quad (9)$$

The bound is sharp.

Proof. First let $\|A\|_1 < 1$ and write $R = (I - A)^{-1} \geq 0$. If an off-diagonal entry R_{ij} attained the maximum b_i of row i , then $R = I + RA$ would imply

$$b_i = \sum_k R_{ik} A_{kj} \leq b_i \sum_k A_{kj} < b_i,$$

since $b_i \geq R_{ii} \geq 1$. Thus each row maximum is diagonal and $\|R\|_1 \leq \text{tr } R$. With eigenvalues λ_j of A counted with algebraic multiplicity,

$$\text{tr } R = \left| \sum_{j=1}^N \frac{1}{1 - \lambda_j} \right| \leq \sum_{j=1}^N \frac{1}{|1 - \lambda_j|} \leq \frac{N}{1 - r}.$$

For column sums at most one, apply the result to tA , $0 < t < 1$, and let $t \uparrow 1$. Invertibility of $I - A$ justifies the limit. To attain the bound, let S be the shift along N sectors and set $A = rI + (1 - r)S$. Then

$$(I - A)^{-1} = (1 - r)^{-1} \sum_{k=0}^{N-1} S^k,$$

whose first column sums to $N/(1 - r)$. Replacing S by θS , $0 \leq \theta < 1$, approaches the bound through strict contractions as $\theta \uparrow 1$. $\square$

For an approximate plan $\hat{x}$, let $u = \hat{x} - d - A\hat{x}$. Since $\hat{x} - x = (I - A)^{-1}u$, (9) gives

$$\|\hat{x} - x\|_1 \leq \frac{N}{1 - r} \|u\|_1. \quad (10)$$

With a fixed spectral gap, residuals of order $o(1/N)$ therefore guarantee convergence. The order cannot be relaxed to $1/N$ as a uniform guarantee: in Proposition 2, $\hat{x}_n = e_1$ has residual norm H_n^{-1} of order $1/n$ and error one. There are $n + 2$ active sectors, including the baseline sector. By (5), the same bound controls sensitivity to perturbations of final demand and technology.

6.3 A common finite-round condition

A loss of intermediate requirements over a common number of rounds supplies the uniform control that the spectral gap lacks.

Proposition 4. Let A_n , $n \geq 0$, be nonnegative bounded operators on ℓ^1 with $\|A_n\|_1 \leq 1$. Suppose there are a common integer $m \geq 1$ and $q \in (0, 1)$ such that

$$\sup_{n \geq 0} \|A_n^m\|_1 \leq q. \quad (11)$$

Then all $I - A_n$ are invertible and

$$\sup_{n \geq 0} \|(I - A_n)^{-1}\|_1 \leq \frac{m}{1 - q}. \quad (12)$$

If additionally $A_n \rightarrow A_0$ strongly and $d_n \rightarrow d_0$ in ℓ^1 , then $x_n = (I - A_n)^{-1}d_n \rightarrow x_0$. For any approximate plan,

$$\|\hat{x}_n - x_n\|_1 \leq \frac{m}{1 - q} \|\hat{x}_n - d_n - A_n \hat{x}_n\|_1. \quad (13)$$

Proof. For $k = jm + s$, $0 \leq s < m$, submultiplicativity gives $\|A_n^k\|_1 \leq q^j$. Hence the Neumann series converges in operator norm and

$$\sum_{k=0}^{\infty} \|A_n^k\|_1 \leq \sum_{j=0}^{\infty} \sum_{s=0}^{m-1} q^j = \frac{m}{1 - q}.$$

This proves (12). Apply it to (5) for continuity and to the residual identity preceding (10) for the error certificate. $\square$

Condition (11) means that at least a fraction $1 - q$ of each sector's initial requirement has left intermediate production within m rounds. It is weaker than a common one-round contraction factor: a two-link chain is extinguished after three rounds even if each input coefficient tends to one. For the affine map $T_n x = d_n + A_n x$, its m th iterate has linear part A_n^m , so this condition controls propagation over several rounds instead of requiring the one-round modulus bound in Theorem 1(i). The growing chains violate it: for every fixed m and all $n \geq m$, $\|A_n^m\|_1 = q_n^m \rightarrow 1$.

For finite input tables, the condition is checked by the largest column sum of A_n^m ; the same m, q must apply across tables. A finite observed block cannot certify unobserved sectors without an additional tail bound. Together with Section 5, this separates existence at each parameter from the control of accumulated effects needed for comparative statics and approximation.

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